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SUBJECT: SE-LAB1

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #45 – Multi-Cloud Asset Cost Optimization Portal

1. Requirements Table

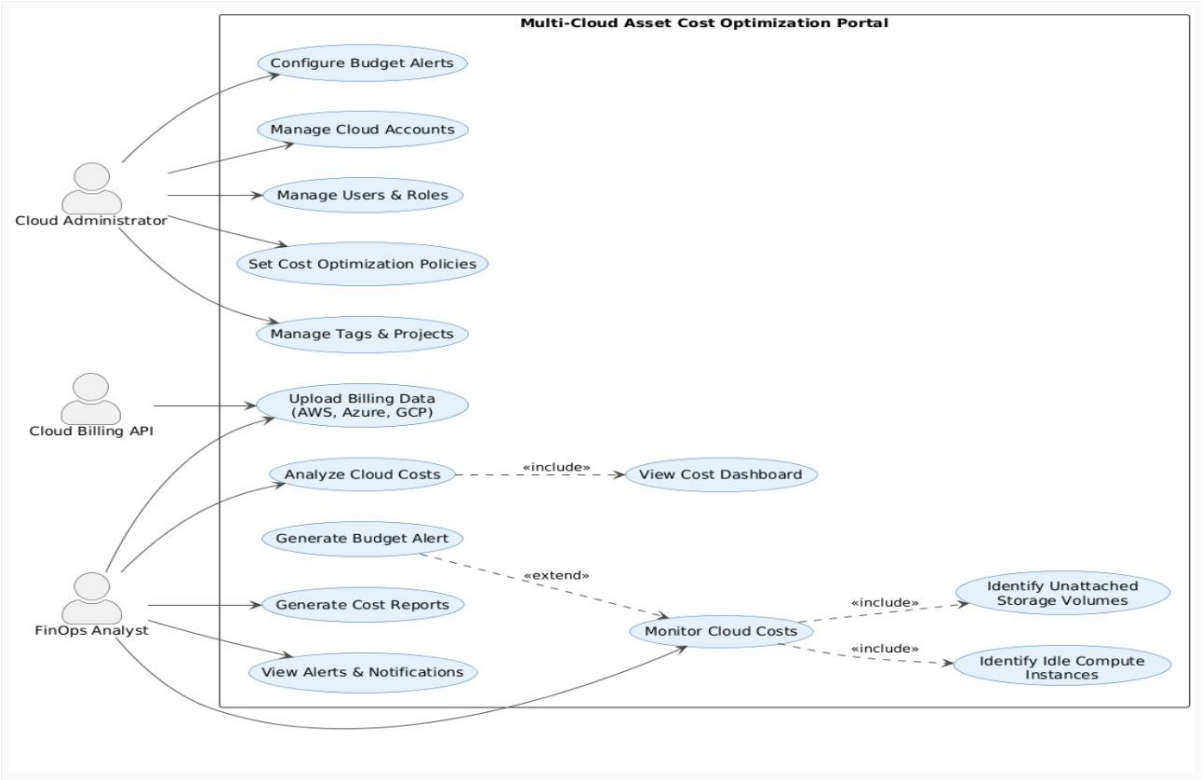
Exactly five Functional Requirements (FR-001 to FR-005) and two Nonfunctional Requirements (NFR-001, NFR-002). FR-001 and NFR-001 are the requirements given in the problem statement; the remaining requirements were derived from the use-case diagram.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest cloud billing CSV exports from AWS, Azure, and GCP, categorize costs by service and project tag, and flag compute instances with less than 5% CPU utilization over a trailing 14-day period.	High	Pass: an idle cloud asset is flagged with an estimated monthly savings figure. Fail: an actively used instance is misclassified as idle.	Core ingestion and idle-detection capability the rest of the portal depends on.
FR-002	Functional	The system shall scan ingested storage billing records and identify storage volumes that are not attached to any active compute instance.	High	Pass: an unattached volume older than 24 hours is listed with its size and estimated monthly cost. Fail: a volume attached to a running instance is listed as unattached.	Unattached storage is a recurring, easily eliminated source of cloud waste for SMB customers.
FR-003	Functional	The system shall generate and dispatch a budget breach alert to the FinOps Analyst and Cloud Administrator whenever actual or projected spend for a project or account exceeds the configured budget threshold.	High	Pass: when spend crosses the configured threshold, an alert is generated within 5 minutes and delivered to all subscribed users. Fail: threshold is crossed and no alert is generated within 5 minutes.	Timely alerting is the primary mechanism for preventing budget overruns.
FR-004	Functional	The system shall allow a FinOps Analyst to generate a cost report, filterable by cloud provider, project tag, and date range, and export it as CSV or PDF.	Medium	Pass: a generated report reflects the selected filters and downloads successfully in the chosen format. Fail: exported figures do not match the filtered dashboard totals.	Analysts need shareable, point-in-time reports for stakeholder reviews and audits.
FR-005	Functional	The system shall allow a Cloud Administrator to add, configure, and remove connected AWS, Azure, and GCP billing accounts, including entry and validation of API credentials.	Medium	Pass: a newly connected account with valid credentials appears as 'Active' and billing data begins ingesting within one billing cycle. Fail: invalid credentials are accepted without an error message.	Multi-cloud coverage depends on administrators being able to onboard and manage account connections.
NFR-001	Nonfunctional	The portal must parse multi-gigabyte cloud billing files and update cost dashboards in under 2 minutes, while meeting baseline security standards for data in transit and at rest.	High	Pass: benchmarking tests confirm dashboard refresh completes within 2 minutes under simulated peak load, with TLS in transit and encryption at rest verified. Fail: refresh exceeds 2 minutes or data is found unencrypted.	Given by the assignment; performance and security are prerequisites for user trust in cost data.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-002	Nonfunctional	The system shall enforce role-based access control so that only users with the Cloud Administrator role can perform account, user, and policy management actions (e.g., Manage Cloud Accounts, Manage Users & Roles, Set Cost Optimization Policies); FinOps Analysts are restricted to monitoring, reporting, and alert-related use cases.	High	Pass: a FinOps Analyst account attempting an administrator-only action is denied with an authorization error. Fail: a FinOps Analyst account is able to complete an administrator-only action.	Separating administrative and analytical permissions limits misconfiguration risk across shared multi-cloud accounts.

2. UML Use-Case Diagram

Actors: Cloud Administrator, Cloud Billing API, FinOps Analyst. The diagram includes an «include» relationship (Analyze Cloud Costs → View Cost Dashboard; Monitor Cloud Costs → Identify Unattached Storage Volumes / Identify Idle Compute Instances) and an «extend» relationship (Generate Budget Alert extends Monitor Cloud Costs).



3. Use-Case Flow Specification

Core use case detailed below: Monitor Cloud Costs, chosen because it demonstrates both the «include» and «extend» relationships shown in the diagram.

Use Case	Monitor Cloud Costs
Primary Actor	FinOps Analyst
Secondary Actors	Cloud Administrator (receives budget breach alerts); Cloud Billing API (source of ingested billing data)
Preconditions	1) The FinOps Analyst is logged into the Multi-Cloud Asset Cost Optimization Portal. 2) Billing data for at least one connected AWS, Azure, or GCP account has already been ingested via Upload Billing Data. 3) Cost optimization policies and budget thresholds have been configured by a Cloud Administrator.
Postconditions	Idle compute instances and unattached storage volumes are identified and listed with estimated savings; if a budget threshold was crossed, a budget alert has been generated and dispatched; the FinOps Analyst has an up-to-date view of cost-saving opportunities.

Main Success Scenario

1. The FinOps Analyst selects “Monitor Cloud Costs” from the portal dashboard.
2. The system retrieves the latest ingested billing data across all connected AWS, Azure, and GCP accounts.
3. The system includes «Identify Idle Compute Instances»: it scans compute billing records and flags instances with less than 5% CPU utilization over the trailing 14 days.
4. The system includes «Identify Unattached Storage Volumes»: it scans storage billing records and flags volumes not attached to any active compute instance.
5. The system aggregates the flagged resources and calculates an estimated monthly savings figure for each.
6. The system checks current and projected spend against the configured budget thresholds for the relevant projects and accounts.
7. The system displays the flagged idle instances, unattached volumes, and spend status on the Monitor Cloud Costs summary view.
8. The FinOps Analyst reviews the flagged resources and spend status.
9. The use case ends successfully.

Alternate Flow – 6a. Budget Threshold Exceeded

6a1. While checking spend in step 6, the system detects that actual or projected spend for a project or account has exceeded its configured budget threshold.

6a2. The system extends the flow with «Generate Budget Alert».

6a3. The system dispatches a budget breach alert to the FinOps Analyst and the Cloud Administrator, including the account, amount over budget, and time detected.

6a4. The FinOps Analyst acknowledges the alert from View Alerts & Notifications.

6a5. The flow resumes at step 7 of the main success scenario.